CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review: 01
Date of Review: 07.08.2026
Project Title: Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators
Cohort: Dr. V. Vaishnavi
Project Guide: Dr. Arun Nimmala
Student Details:

S.No.	Student Name	Student ID	Signature
1	Velivala Keerthi	**2320040083**	
2	T. Venkata Ram Charan Reddy	**2320040017**	
3	Shaik Hanifa	**2320040009**	
4	K. Dinesh	**2320040095**	

Brief Description of the Project

Convolutional Neural Network (CNN) edge accelerators are constrained by the high area and power dissipation of Multiply-Accumulate (MAC) datapaths, where 8x8 exact multipliers dominate silicon cost (68 LUTs on Xilinx Zynq-7020). While approximate computing trades controlled arithmetic error for hardware savings, uniform approximation causes severe accuracy degradation or negligible gains. Existing layer-wise mapping methods rely on computationally prohibitive evolutionary search (ALWANN: 7.5 CPU-days) or GPU retraining loops (TRAM). This project proposes a lightweight, single-pass sensitivity metric computed in 0.2 seconds combining static weight variance and dynamic loss gradients via a 256-sample calibration probe without retraining. Layers are ranked such that error-vulnerable early layers receive exact multipliers while noise-resilient deep layers receive aggressive approximate multipliers. The heterogeneous design is mapped onto synthesizable Verilog Processing Elements (PEs) and deployed on a PYNQ-Z2 FPGA board via AXI4-Stream DMA, achieving up to 32.5% multiplier area reduction while preserving baseline classification accuracy.

Progress Achieved up to This Review

- **Multiplier RTL Library Design:** Designed 4 distinct 8x8 multiplier architectures in synthesizable Verilog (exact_mult8x8, trunc_mult8x8, compensated_mult8x8, and orbased_mult8x8) parameterized by truncation depth T.
- **Exhaustive Simulation & Error Characterization:** Developed exhaustive 65,536-vector testbenches in Icarus Verilog; extracted exact error metrics (Mean Error, NMED, MRED, Max Error) across all truncation depths (T=2 to T=6).
- **Vivado Synthesis & Circuit Dominance:** Synthesized on Xilinx Zynq-7020 (xc7z020). Demonstrated that Compensated strictly dominates Truncated at identical 56 LUTs (-17.6%) with 35% lower error via zero-cost +2^(T-1) carry-in folding; OR-based discarded.
- **Theoretical Sensitivity Formulations:** Formulated 5 candidate layer-sensitivity metrics (S1 to S5) combining static weight tensor variance with backpropagated loss gradient norms.
- **Benchmark Protocols & Decision Gate:** Established baseline evaluation criteria on Fashion-MNIST, test protocols, and defined the strict Spearman rank correlation clearance gate (|ρ| >= 0.70) required prior to hardware datapath commitment.

Signature of Project Guide: _____ **Date:** _____